

Introduction to ABAP Core Data Services

ABAP Platform Product Management, SAP SE
May 2024

PUBLIC

Agenda

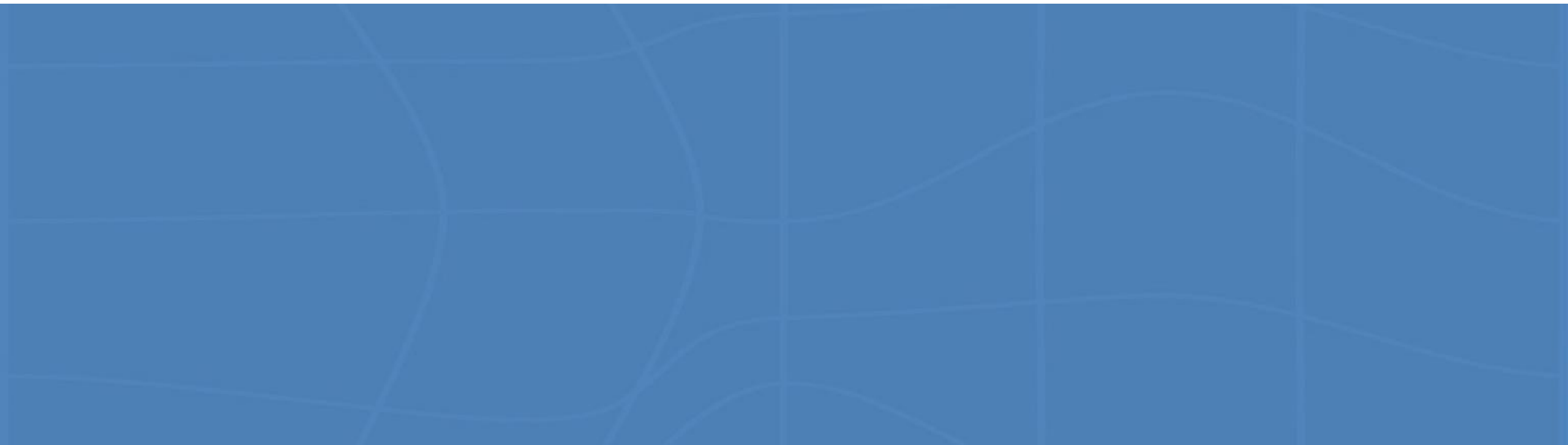
MODERN ABAP DEVELOPMENT

ABAP CORE DATA SERVICES (CDS) OVERVIEW

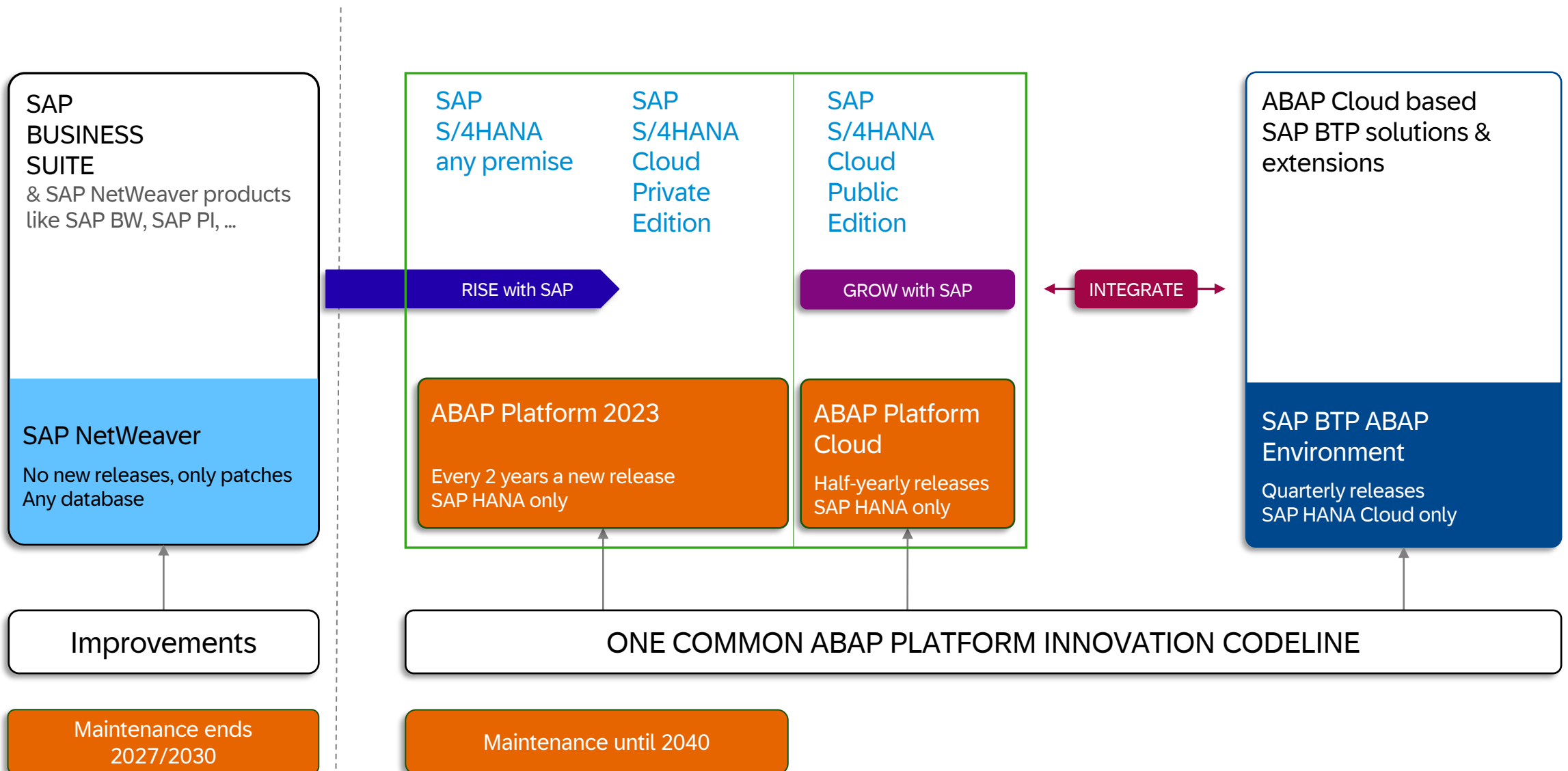
CDS IN ABAP APPLICATION PROGRAMMING

SUMMARY

MODERN ABAP DEVELOPMENT

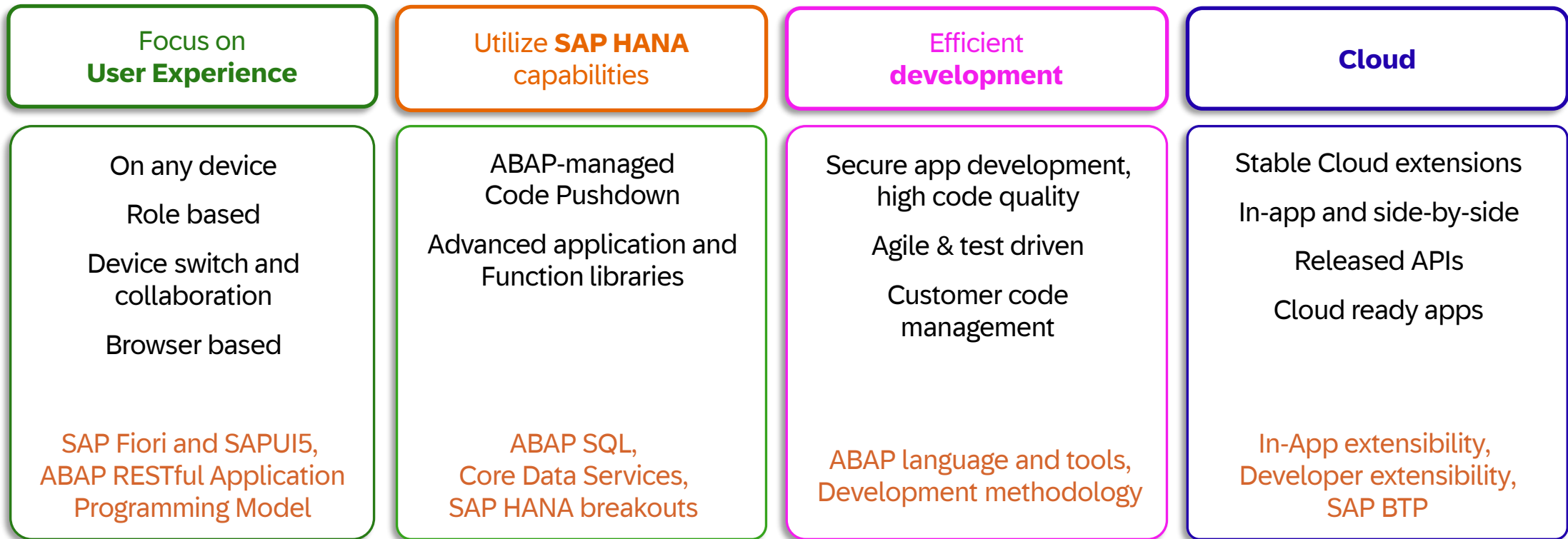


ABAP Platform – Foundation of SAP's ERP applications



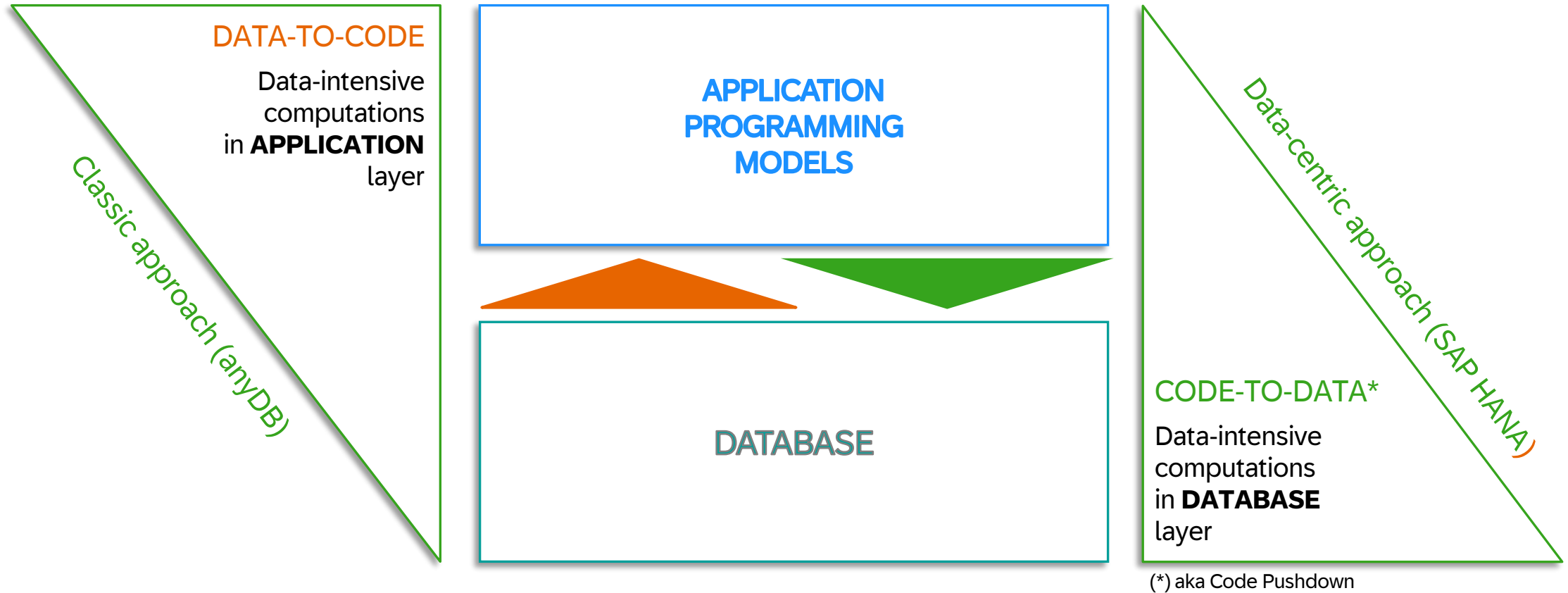
Modern ABAP application development

Must learn topics: UX, SAP HANA, state of the art development, Cloud



Modern ABAP application development

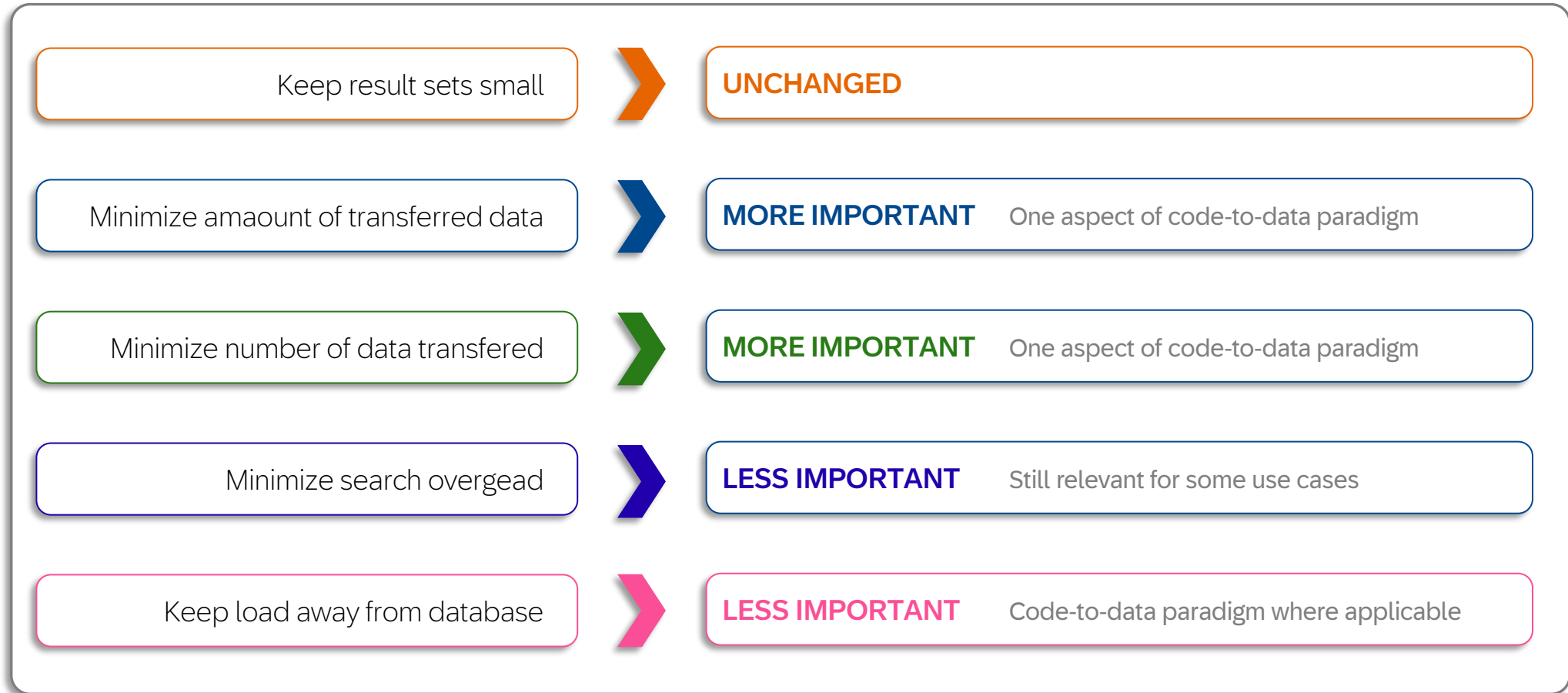
Programming paradigm shift powered by SAP HANA



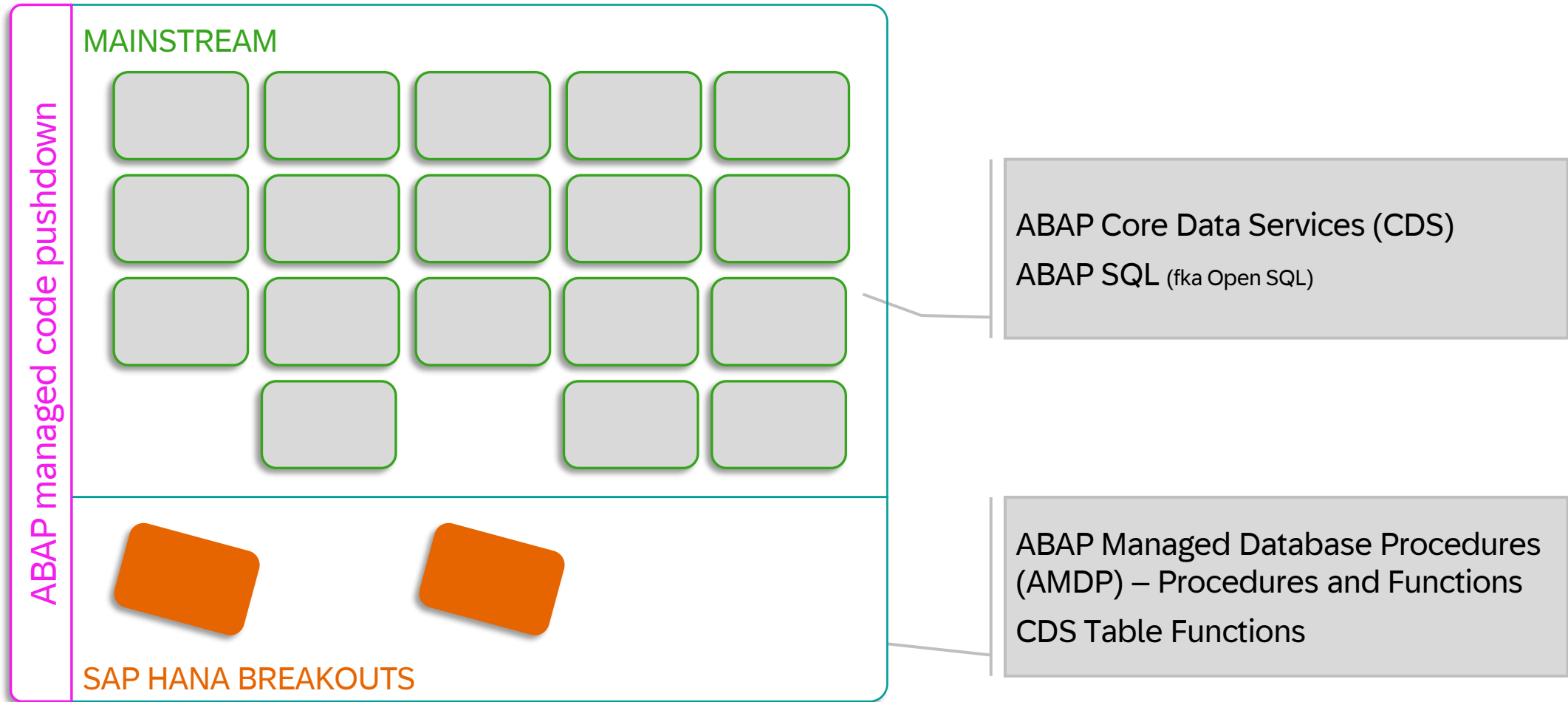
More in this [blog](#)

Modern ABAP application development

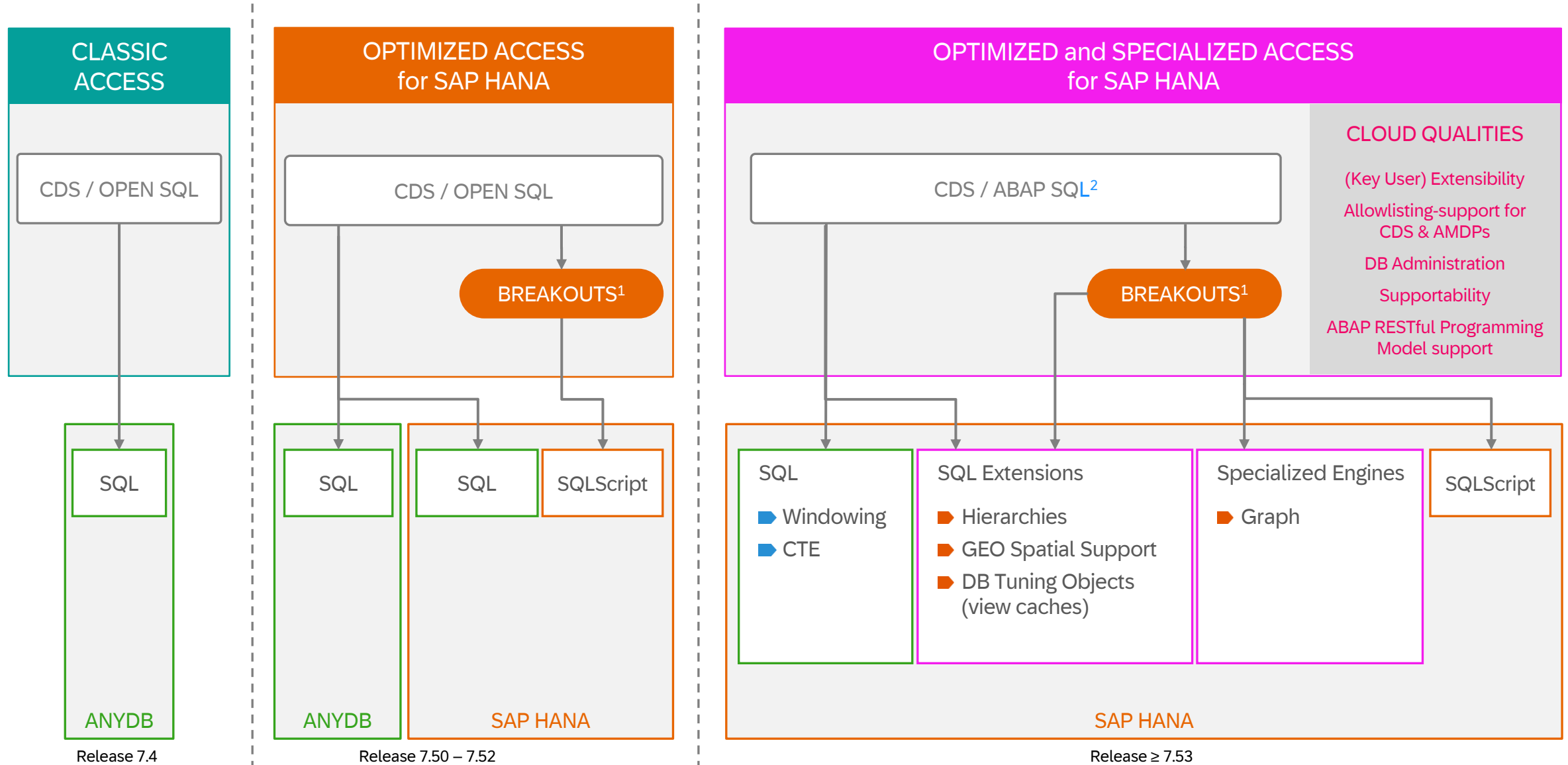
Golden rules for SQL programming – Priority shift on SAP HANA



Modern ABAP application development on SAP HANA



Evolution of ABAP CDS and Open/ABAP SQL



¹ AMDP procedures and functions, CDS table functions, and CDS hierarchies

² Open SQL was renamed to ABAP SQL starting with ABAP Platform 1809

Efficient ABAP development in Eclipse

MODERN DEVELOPMENT TOOLSET

Fully eclipse-based
Syntax check, code completion
Navigation, search, quick fixes
Full support for the ABAP RESTful
Application Programming Model and CDS

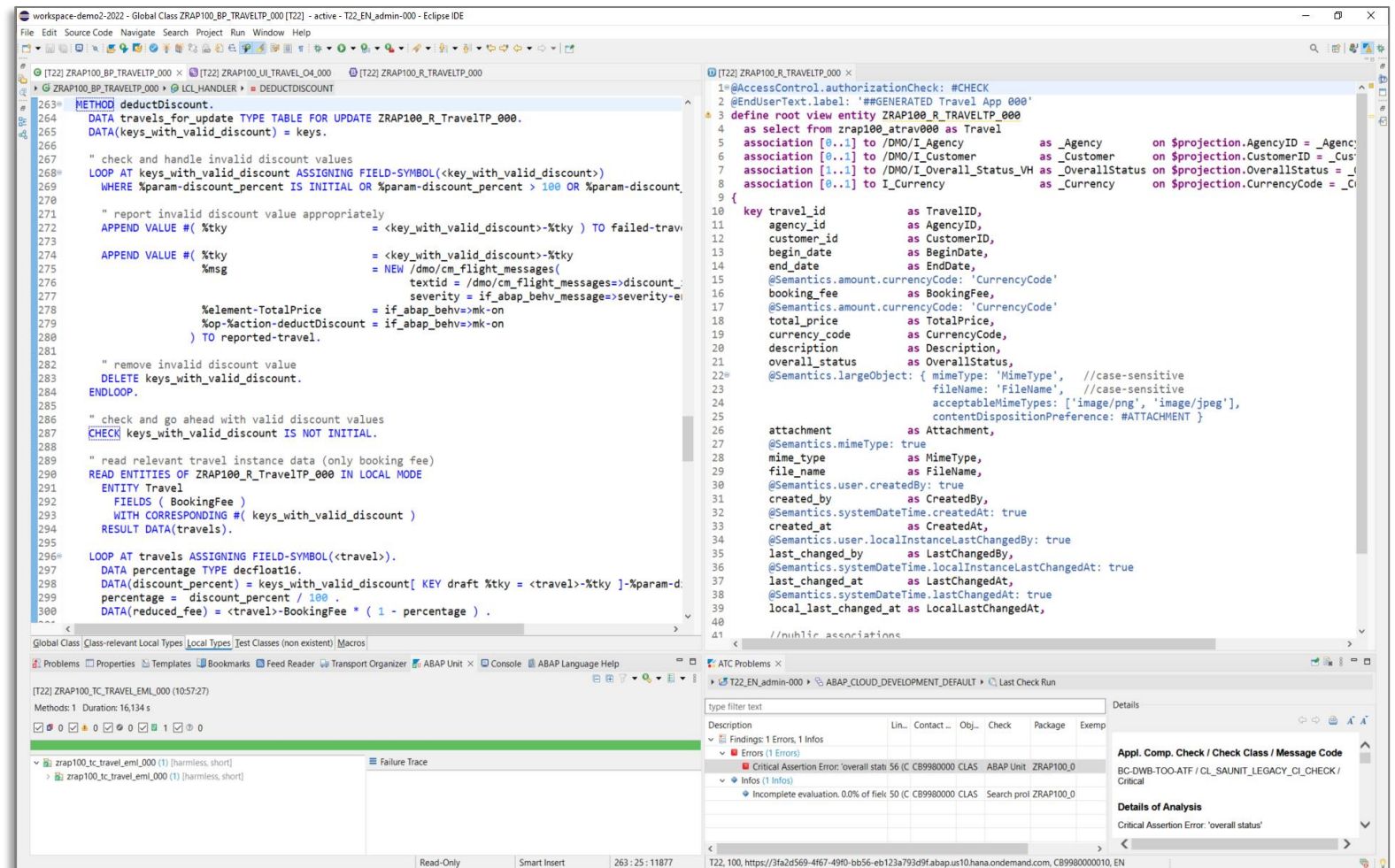
QUALITY ASSURANCE

Static code checks (ATC, CVA) with
remote and local scenarios
Unit testing incl. isolation frameworks
Test seams and injections

SUPPORTABILITY

Debugging, profiling, tracing
Static and dynamic logging
Runtime monitoring and analysis

HIGH DEVELOPER PRODUCTIVITY ON-PREMISE AND IN THE CLOUD



ABAP CDS OVERVIEW



ABAP CDS – Next generation data modeling and access

COMMON DATA MODEL

Semantically rich
Declarative
Close to conceptual thinking

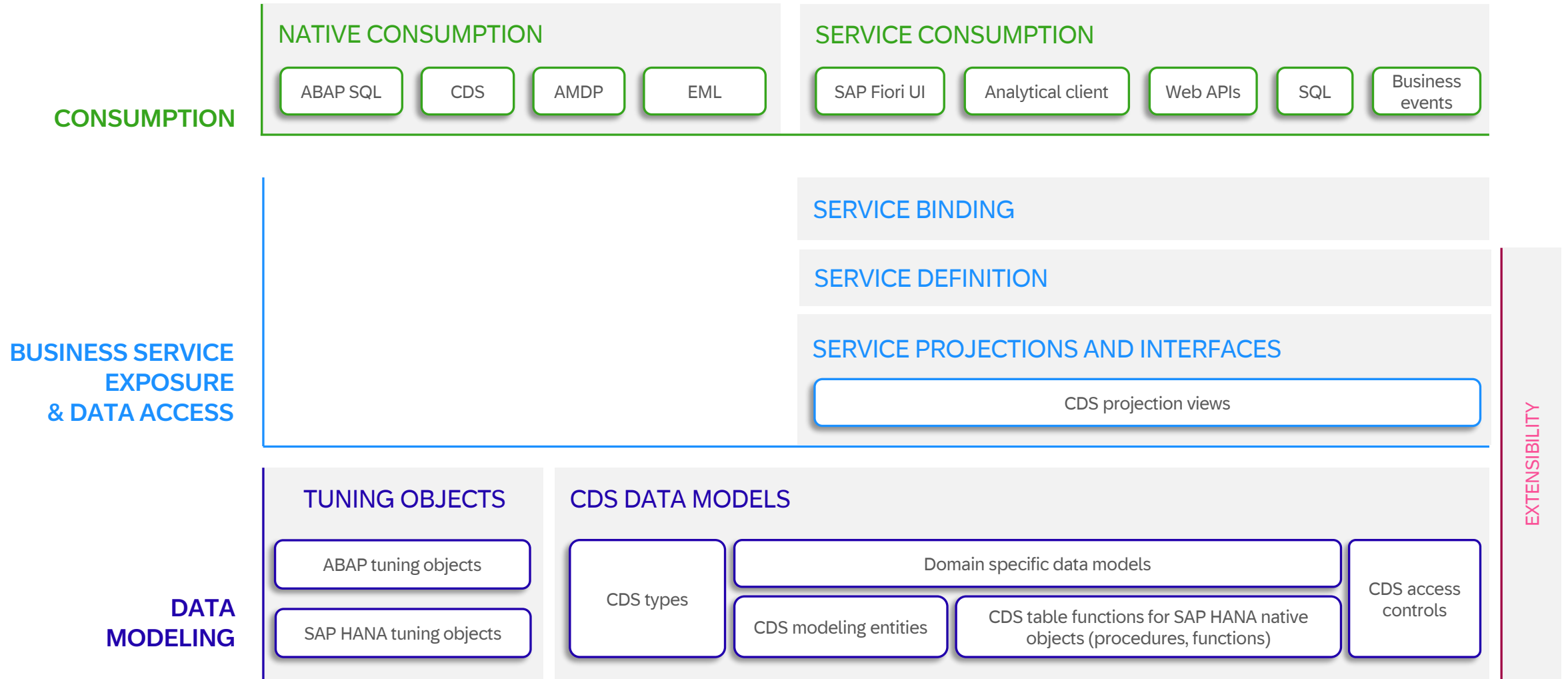
IMPROVED PROGRAMMING MODEL

For all application domains
Rich set of built-in functions
Code pushdown capabilities
Declarative access control

CAPTURE BUSINESS INTENT

Reduced complexity
Extending SQL

ABAP CDS data models



ABAP CDS – Overview of supported CDS entity types¹

Category	ABAP CDS Entity Type	ABAP CDS Statement
Standard view building	CDS View Entity CDS Projection View CDS Projection View – Analytical Query CDS DDIC-based View <i>(deprecated)</i>	DEFINE VIEW ENTITY DEFINE VIEW ENTITY AS PROJECTION ON DEFINE TRANSIENT VIEW ENTITY AS PROJECTION ON DEFINE VIEW <i>(deprecated)</i>
Advanced view building – External implementation	CDS Custom Entity CDS Abstract Entity	DEFINE CUSTOM ENTITY DEFINE ABSTRACT ENTITY
Advanced view building – SAP HANA breakout	CDS Table Function CDS Hierarchy CDS Scalar Function ²	DEFINE TABLE FUNCTION DEFINE HIERARCHY DEFINE SCALAR FUNCTION
Type definition	CDS Simple Type ² CDS Enumerated Type ²	DEFINE SIMPLE TYPE DEFINE TYPE ENUM
Transactional behavior definition of business objects built with the ABAP RESTful Application Programming Model	CDS Behavior Definition (and Projection)	DEFINE BEHAVIOR
Modification-free extension	CDS View Entity Extension CDS Custom Entity Extension CDS Abstract Entity Extension CDS DDIC-based View Extension <i>(deprecated)</i> CDS Metadata Extension CDS Behavior Extension	EXTEND VIEW ENTITY EXTEND CUSTOM ENTITY EXTEND ABSTRACT ENTITY EXTEND VIEW <i>(deprecated)</i> ANNOTATE VIEW EXTEND BEHAVIOR
Access control definition	CDS Access Control	DEFINE ROLE

ABAP CDS view entity – Example

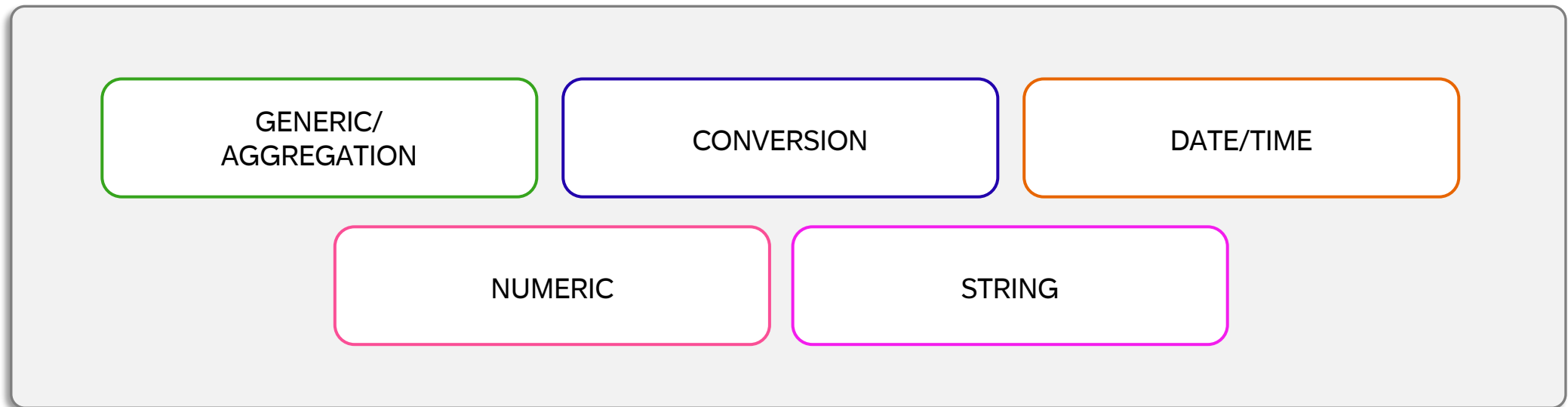
The diagram illustrates the structure of an ABAP CDS view entity, with callouts pointing to specific parts of the code:

- View annotations:** Points to the `@EndUserText.label` and `@AccessControl.authorizationCheck` annotations at the top of the code.
- View definition:** Points to the `define root view entity` statement.
- Associations:** Points to the `association` statements defining relationships between the view and other entities.
- Selection:** Points to the `key` statement defining the primary key of the view.
- Element annotations:** Points to the `@Semantics` annotations on the `amount` and `currencyCode` elements.

```
1 @EndUserText.label: 'Travel View Entity for Draft RefScen'
2 @AccessControl.authorizationCheck: #NOT_REQUIRED
3
4 define root view entity /DMO/I_Travel_D
5   as select from /dmo/a_travel_d
6
7   composition [0..*] of /DMO/I_Booking_D      as _Booking
8
9   association [0..1] to /DMO/I_Agency           as _Agency      on $projection.AgencyID = _Agency.AgencyID
10  association [0..1] to /DMO/I_Customer         as _Customer      on $projection.CustomerID = _Customer.CustomerID
11  association [1..1] to /DMO/I_Overall_Status_VH as _OverallStatus on $projection.OverallStatus = _OverallStatus.OverallStatus
12  association [0..1] to I_Currency              as _Currency      on $projection.CurrencyCode = _Currency.Currency
13
14 { //dmo/a_travel_d
15   key travel_uuid          as TravelUUID,
16
17   travel_id               as TravelID,
18   agency_id              as AgencyID,
19   customer_id            as CustomerID,
20   begin_date             as BeginDate,
21   end_date               as EndDate,
22   @Semantics.amount.currencyCode: 'CurrencyCode'
23   booking_fee            as BookingFee,
24   @Semantics.amount.currencyCode: 'CurrencyCode'
25   total_price            as TotalPrice,
26   currency_code          as CurrencyCode,
27   description            as Description,
28   overall_status         as OverallStatus,
29
30   //total flag field
31   @Semantics.systemDateTime.lastChangedAt: true
32   last_changed_at        as LastChangedAt,
33
34   //Associations
35   _Booking,
36
37   _Agency,
38   _Customer,
39   _OverallStatus,
40   _Currency
41 }
```

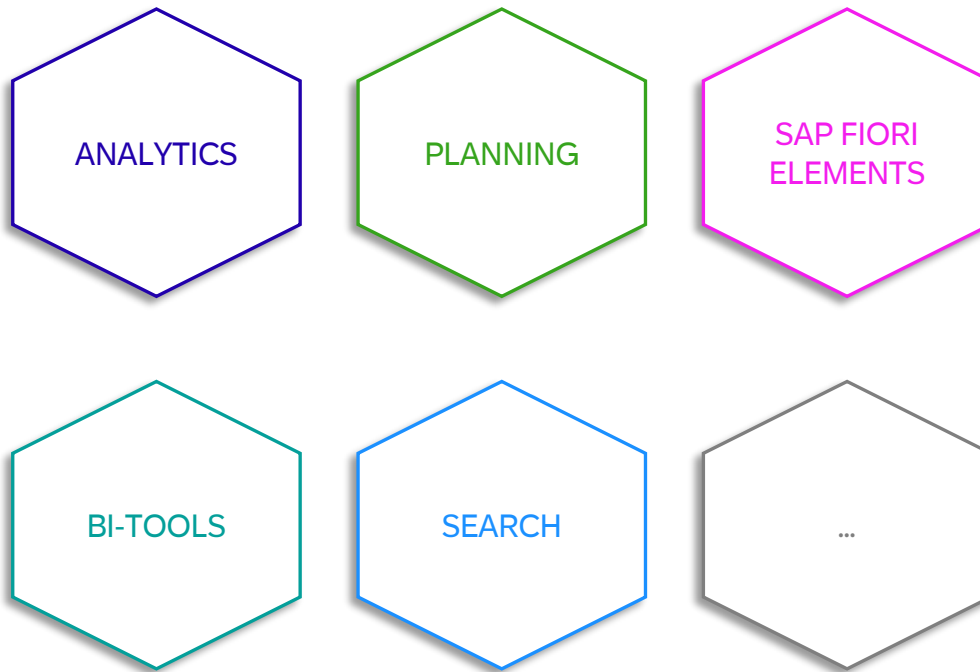
[ABAP Flight Reference Scenario](#) -
Example available in package /DMO/FLIGHT_DRAFT

Built-in SQL functions and expressions



CDS annotations for domain-specific frameworks

CDS
ANNOTATION
DOMAINS



Enrich meta model with consumer-specific vocabularies

Reusable and unified view model for all use cases

Enabling flexible usage in different contexts

Separation of concern using CDS metadata extensions for UI-specific annotations

Vital part of new ABAP programming models

CDS associations

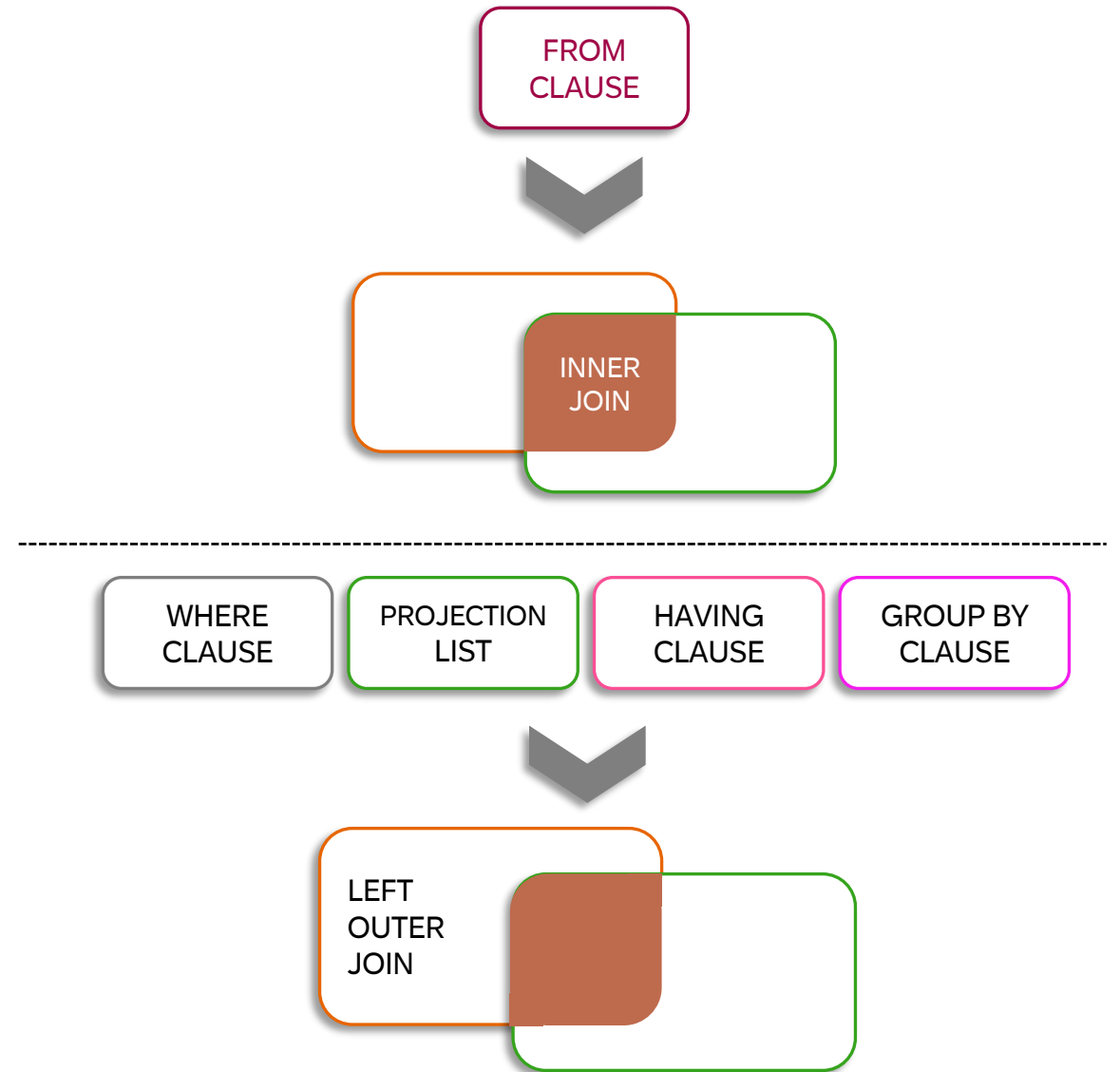
Capturing the relationships between entities in the data model

Used associations are implicitly translated into SQL joins

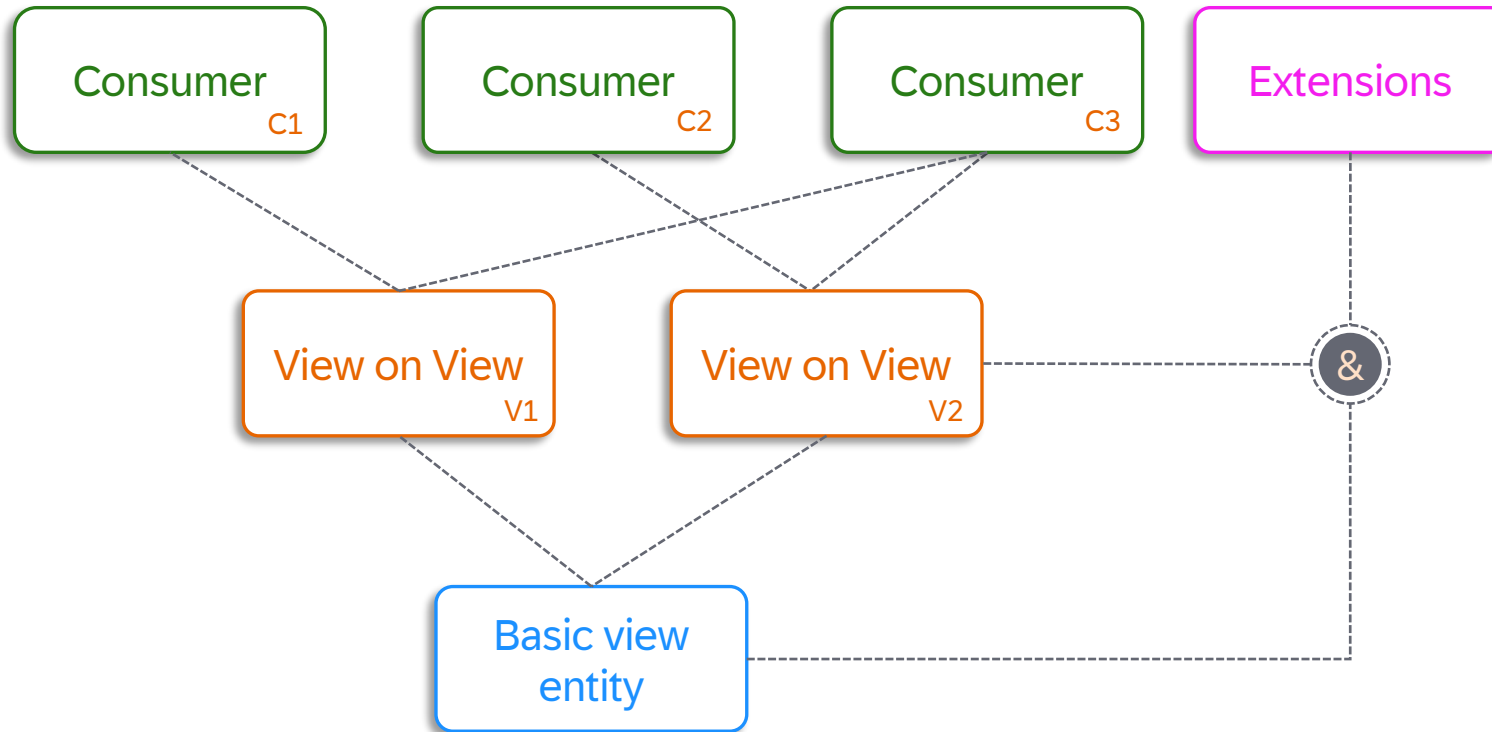
Reuse of generated joins when semantically identical

Can be queried in CDS views and ABAP SQL

▶ Navigate thru entity using path expressions



Nested CDS views and extensibility



Hierarchical view-on-view concept
Optimized result sets with minimum data transfer
Modification-free append of additional columns, arithmetic expressions or literals to result set

CDS hierarchies – Build hierarchies on SAP HANA

CDS Hierarchy defines a CDS entity as a hierarchy in the CDS DDL

A CDS hierarchy has a tabular results set whose rows construct parent-child relationships from hierarchical data sets

A CDS hierarchy can be accessed as the data source of an ABAP SQL query, and it is handled like a hierarchy in which additional hierarchy columns can be selected

```
define view CDS_SOURCE_VIEW
  as select from hierarchy_table_sp
  association [1..*] to CDS_SOURCE_VIEW as _Parent
  //child to parent association (Mandatory)
  on $projection.manager = _Parent.employee
  {
    key employee,
    key manager,

    _Parent // Make association public
  }
```

CDS source view

Self association

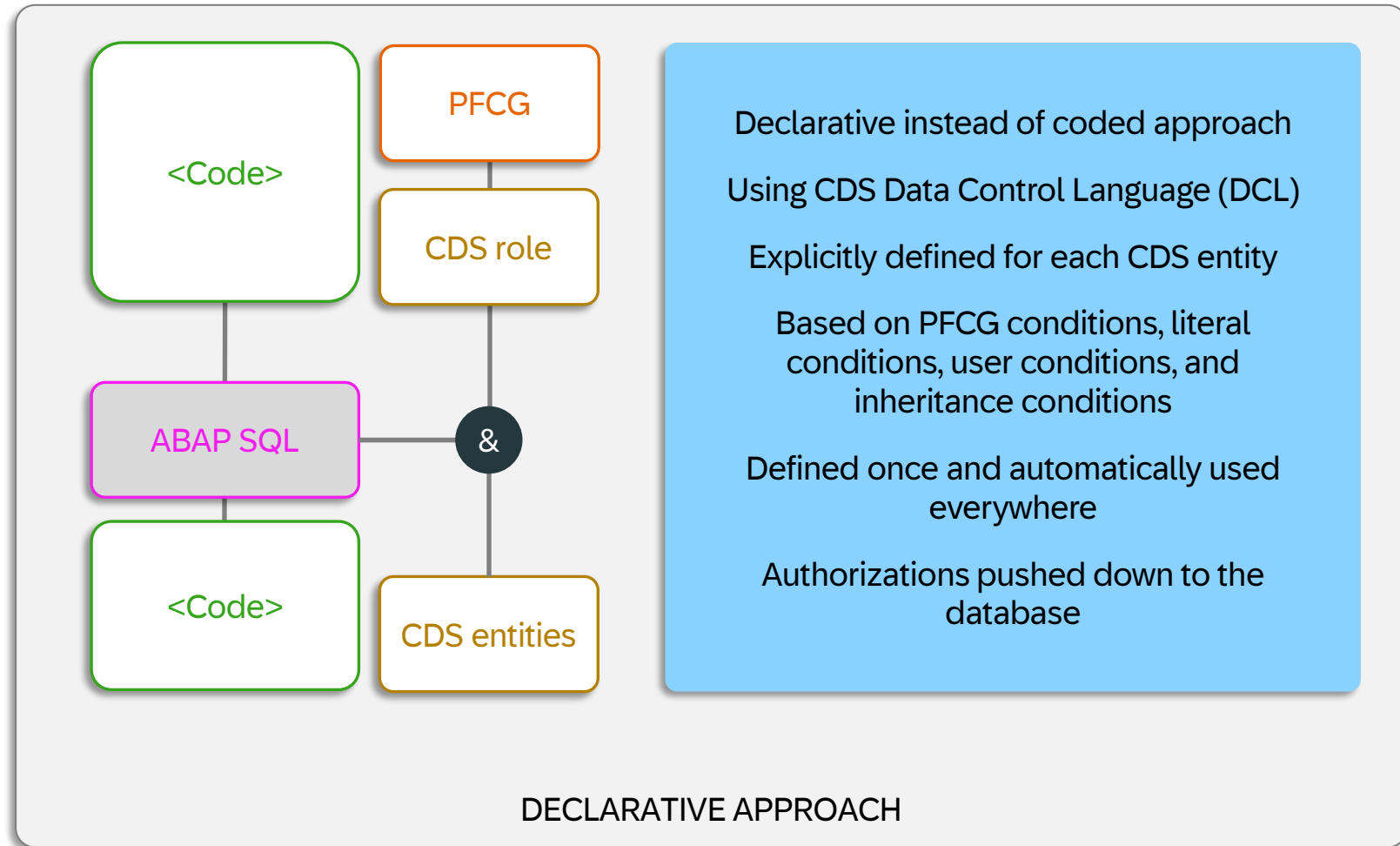
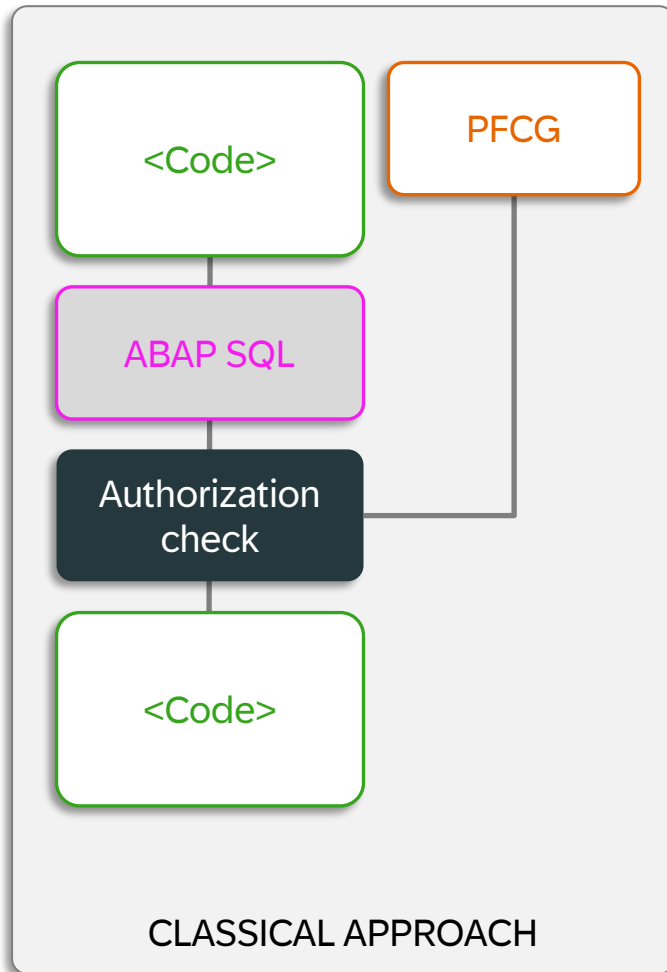
```
define hierarchy CDS_HIERARCHY
  as parent child hierarchy(
    source CDS_SOURCE_VIEW
    child to parent association _Parent
    start where
      SOURCE_CDS_VIEW.cost_center = 100
    siblings order by
      SOURCE_CDS_VIEW.employee
    multiple parents not allowed
  )
  {
    key employee,
    key manager,
    _Parent
  }
```

ABAP CDS hierarchy

CDS source view

Filter results

Declarative access control with CDS roles



CDS development in Eclipse

01

Eclipse-based CDS Tooling
Source-based editor
Part of ABAP Development tools

02

Development support

Syntax check
Code completion
Syntax highlighting, pretty printing
Element info and navigation
Quick fixes

03

Supportability

Enhanced data preview
Dependency analyser
Activation graph / Dictionary log
Active annotations view
Unit testing

SAP HANA Breakouts

Why code breakouts for SAP HANA make sense

UNLEASH THE FULL POTENTIAL OF YOUR SAP HANA DATABASE

Some scenarios require
selective measures

Highest performance requirements
e.g. with complex calculations

Use of database/analytical engine
specific functions required

ABAP SQL and plain CDS views are not sufficient to solve problem efficiently

ABAP managed database procedures (AMDP) for SAP HANA

UTILIZE NATIVE SAP HANA ENTITIES

Stored procedures, database functions and scalar functions supported

Expression of complex logic

Parameterized requests and multiple result sets

INTEGRATED IN THE ABAP INFRASTRUCTURE

Development, runtime error analysis, enhancement, transport

SAP HANA SQLScript coding embedded in ABAP classes

Seamless integration into CDS via CDS table functions

EASY ACCESS TO SAP HANA ADVANCED ENGINES / LIBRARIES

Like predictive analysis, financials, text mining, calculation engine

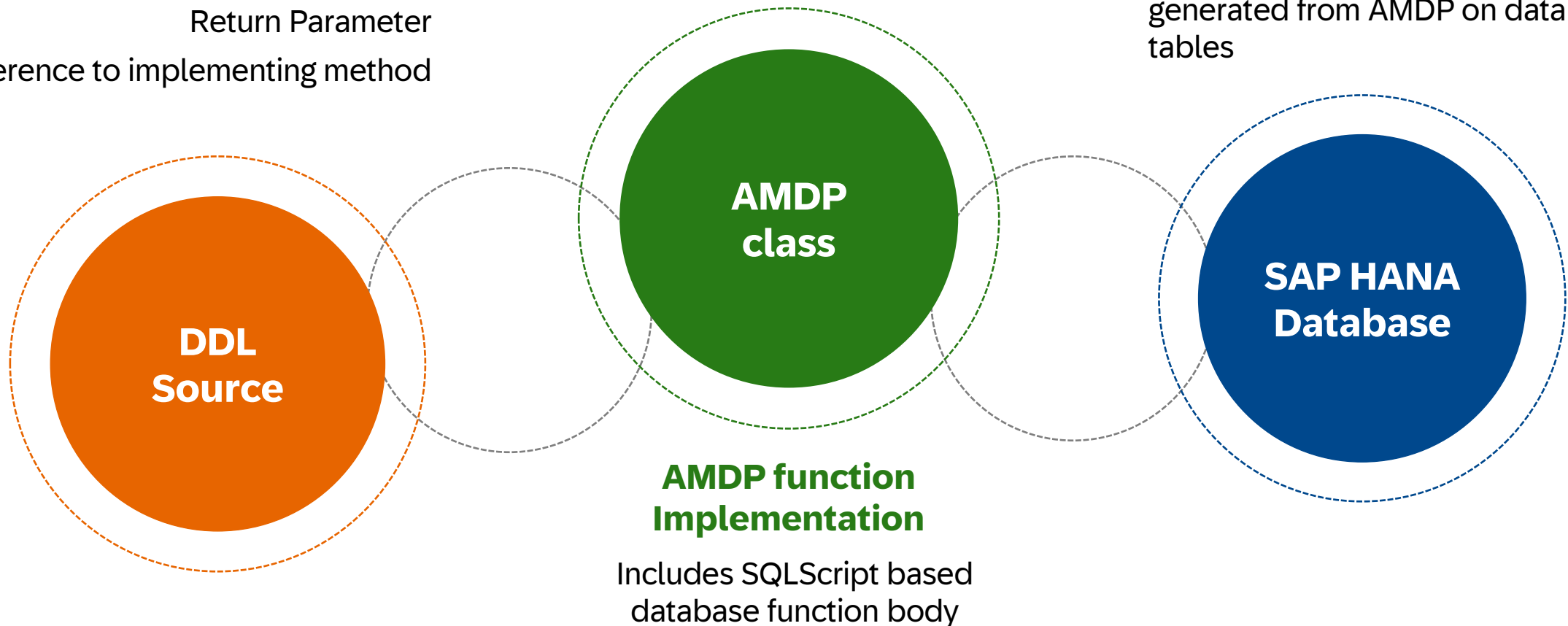
CDS table functions – Seamless AMDP integration into CDS

CDS table function definition

Parameter list
Return Parameter
Reference to implementing method

Runtime for table function

Runs stored SQLScript procedure generated from AMDP on database tables



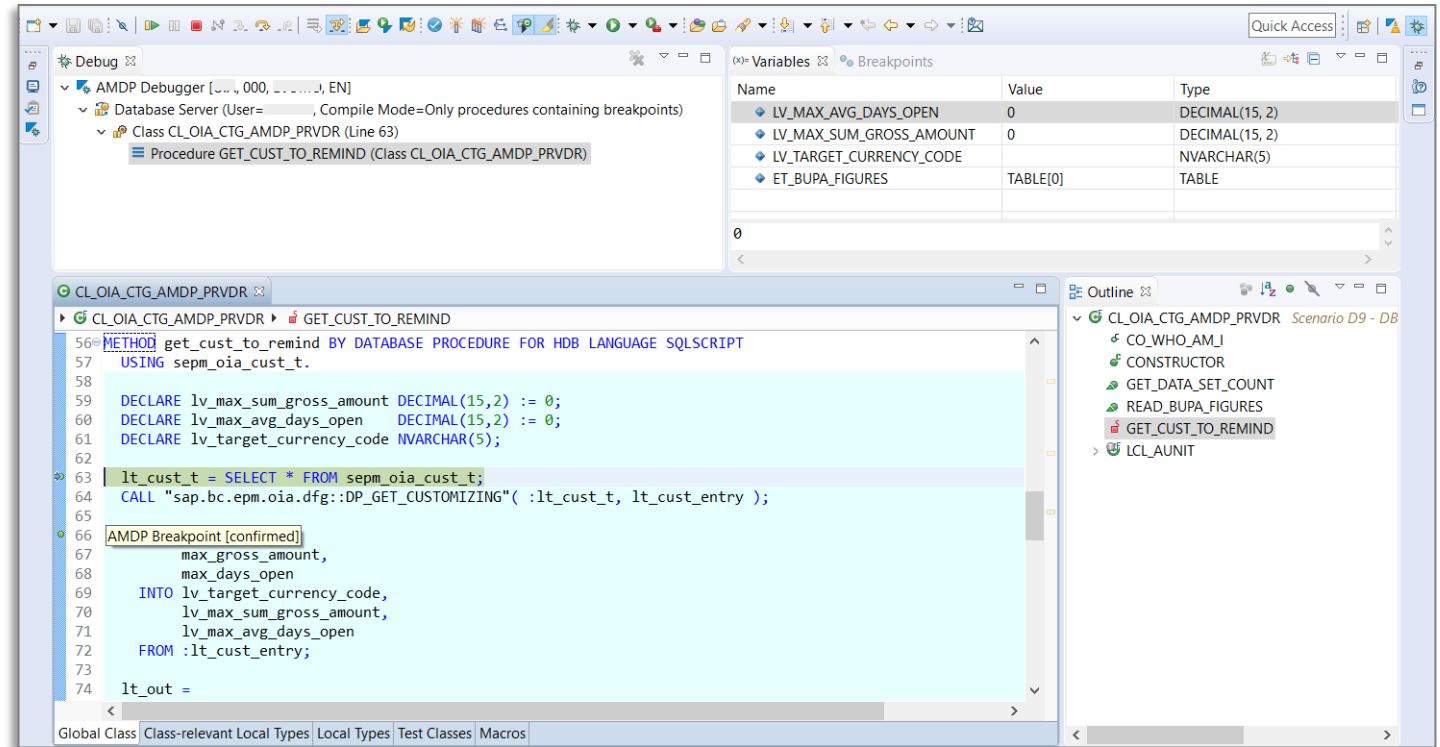
AMDP development in Eclipse

DEVELOPMENT

Eclipse-based source code editor
SQLScript syntax highlighting
Static syntax check
Error handling via class-based exceptions
Modification-free extension via AMDP BADIs

SUPPORTABILITY

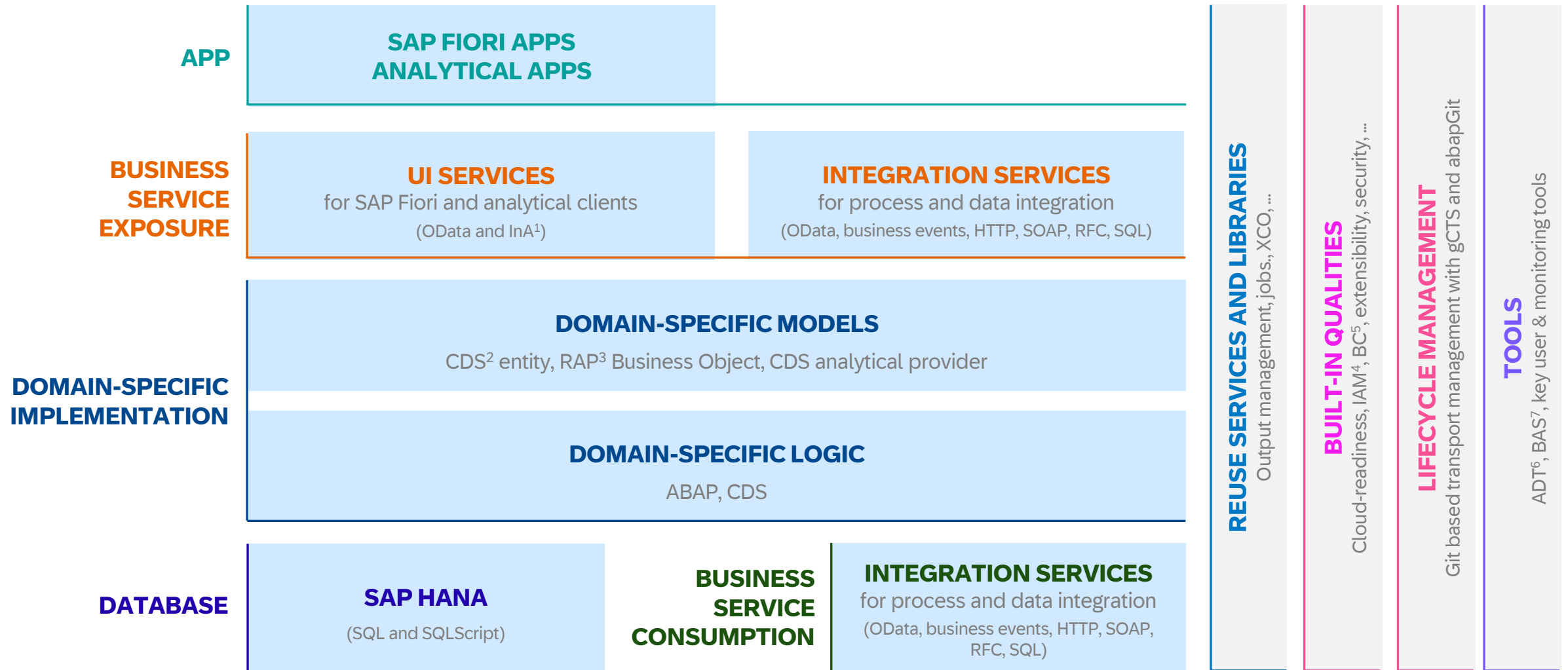
Autonomous AMDP debugger in ADT
Detailed runtime errors integrated in ST22 dumps



CORE DATA SERVICES IN ABAP APPLICATION PROGRAMMING



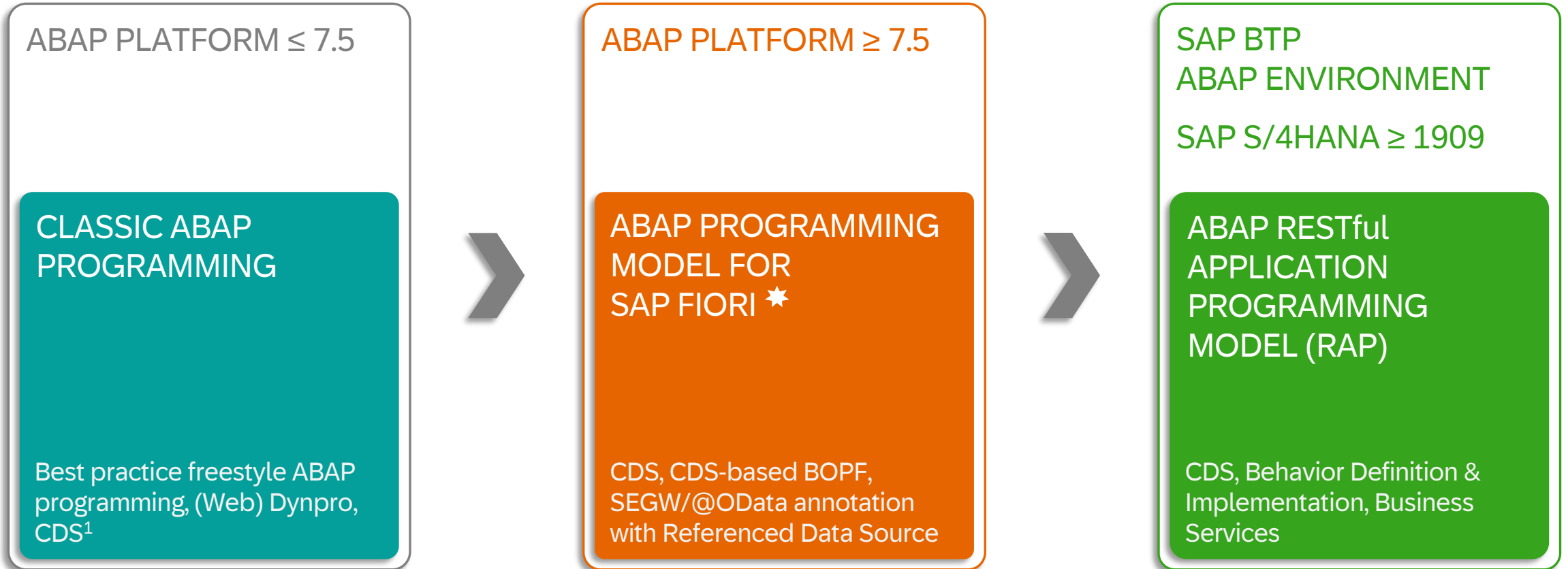
ABAP Cloud big picture



ABAP RESTful APPLICATION PROGRAMMING MODEL (RAP)

Evolution of the ABAP programming model

CDS as vital part of SAP S/4HANA and SAP BTP ABAP Environment

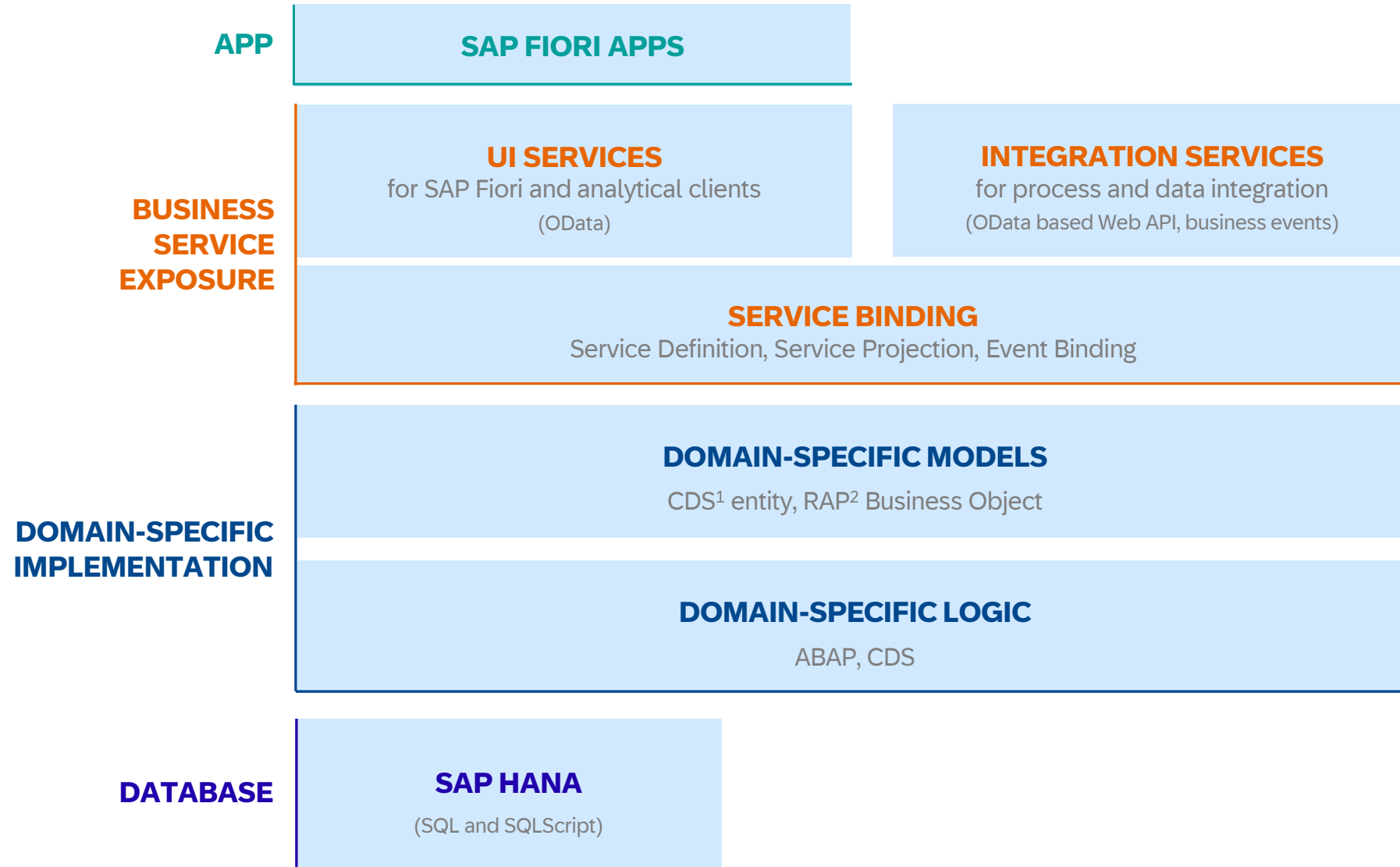


¹ starting with release 7.4 SPS05

★ Safe investments!

ABAP RESTful application programming model – Big picture

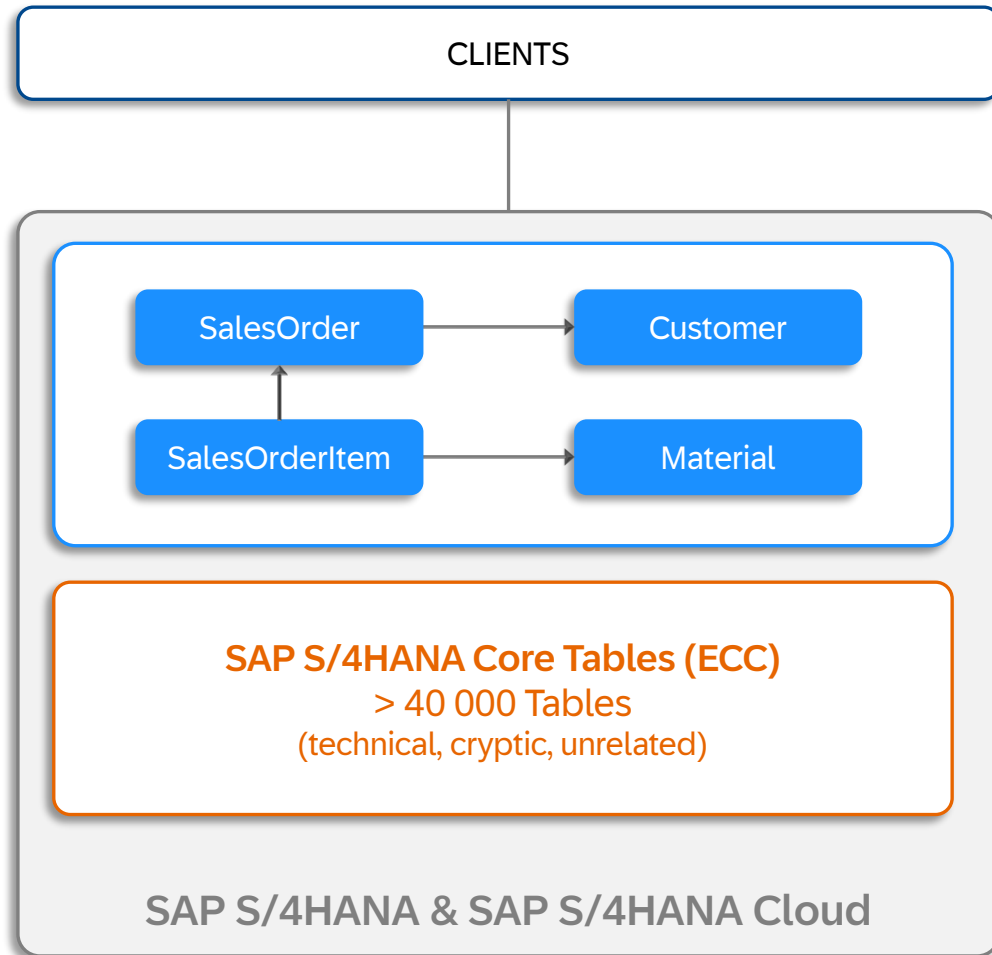
At the heart of ABAP Cloud for transactional scenarios



¹ Core Data Services

² ABAP RESTful application programming model

Virtual Data Model (VDM) for application data in SAP S/4HANA



VDM is a set of views on SAP S/4HANA application data

- Business-oriented
- Understandable & semantically rich
- Reusable & stable
- Executed on SAP HANA
- Implemented as ABAP CDS views

VDM is the data model and source for all types of apps

- Transactional
- Analytical
- External interfaces

Customers and partners can develop on released VDM views

CDS in SAP S/4HANA – Recommendations

- When implementing your own CDS views, follow the [SAP S/4HANA's Virtual Data Model \(VDM\) implementation pattern](#) for best reuse and interoperability
- Clarify your data model before starting your implementation
- Start with the implementation of basic interface views
Adapt names and add semantical information already on this level

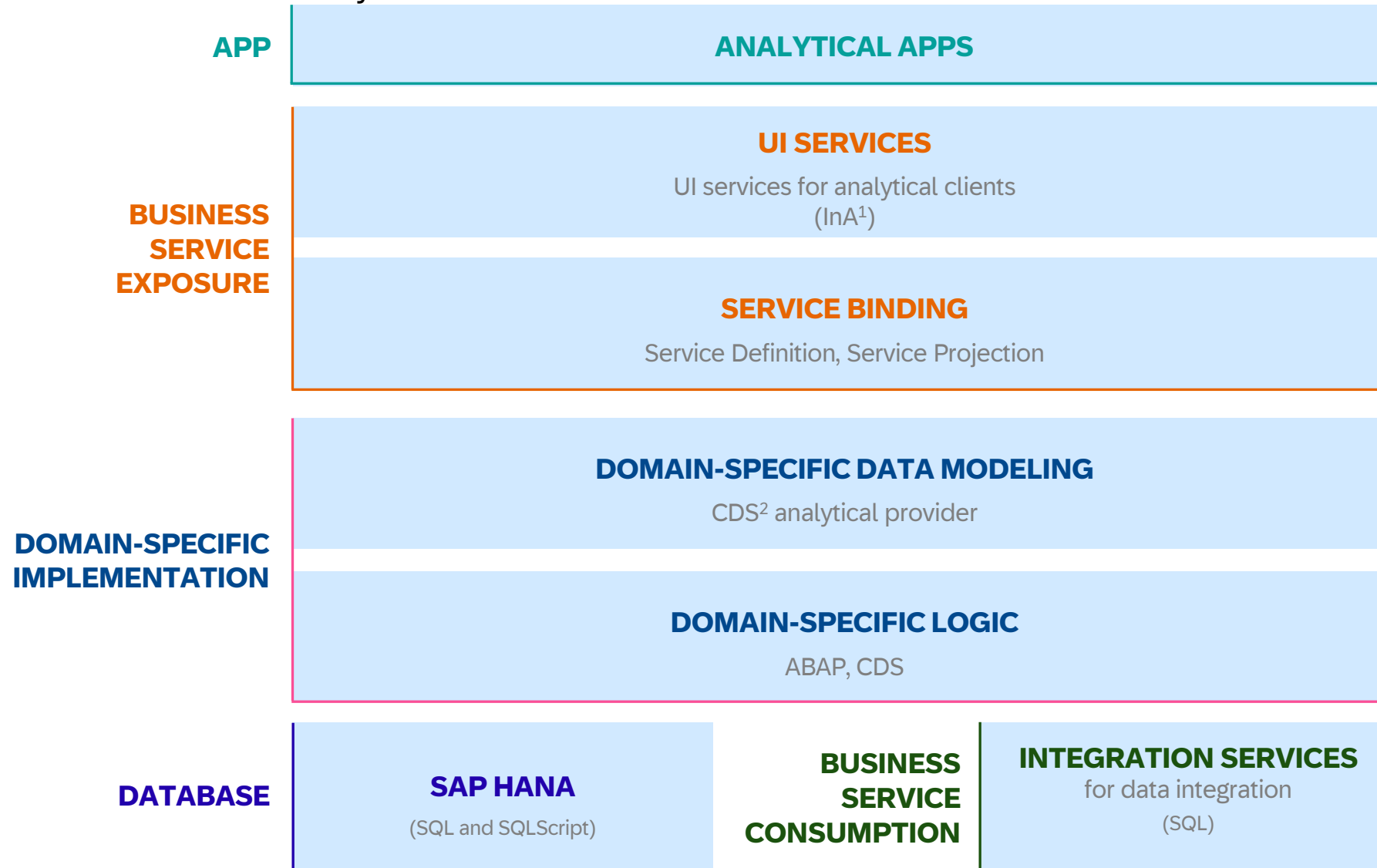
ANALYTICS

Motivation for embedded analytics

	Transactional	Simple Analytics	Multi Dimensional	Embedded Analytics
Intention	Daily business tasks	Fast and easy insights	Interactive analysis	Daily business tasks Fast insights & access to interactive analysis
Interactivity	Linear	Static	Highly dynamic	On demand
Complexity	Low Single data sets	Low-medium Aggregating single sets	High Lots of data sets	Mixed
Aggregations	Very few	Yes	Dynamic: different types on different levels	All
Requests	Simple requests Few data	One request Complexity varies	Each interaction may trigger a complex request	Both
Request Types	CRUD	Read	Read	Both
Data Sources	Live data	Live data May be precalculated	Replicated data & live data	Live data

ABAP analytics – Big picture

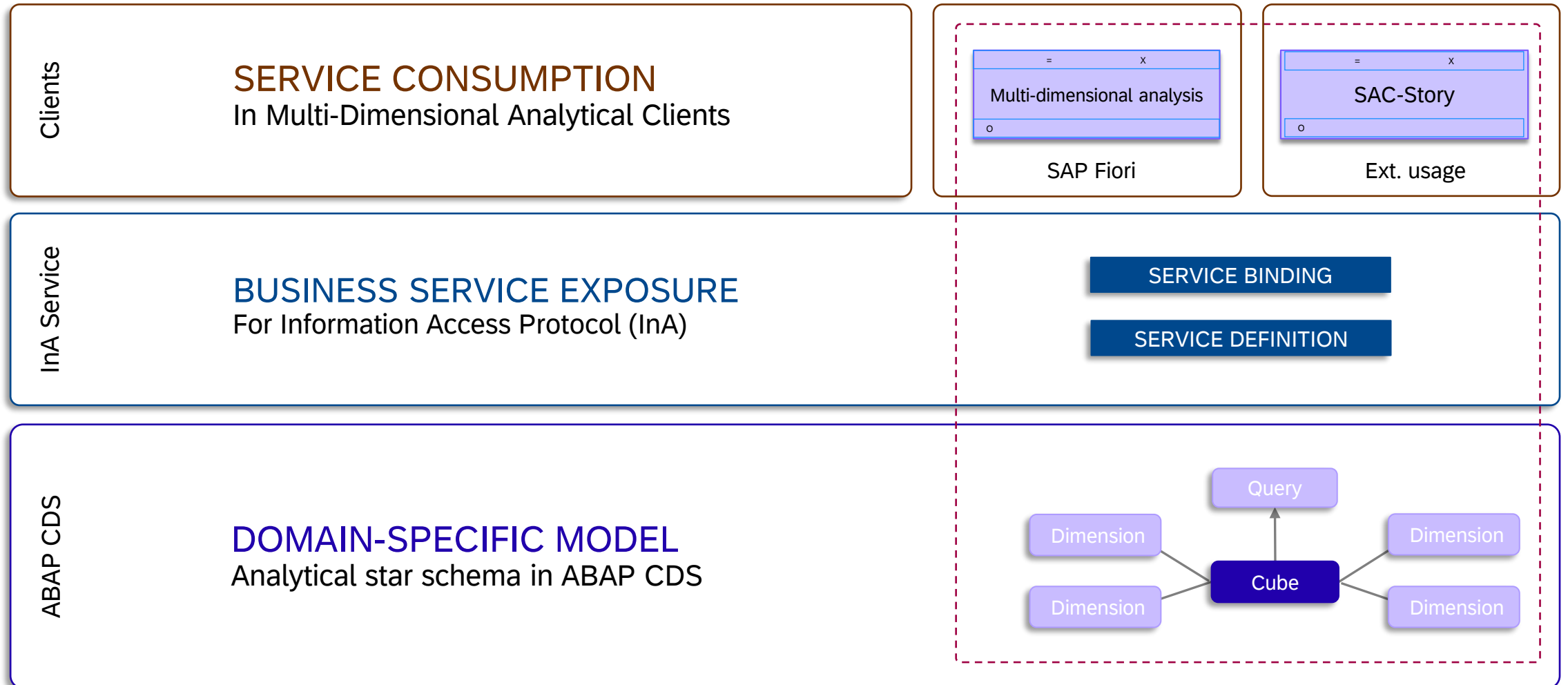
At the heart of ABAP Cloud for analytical scenarios



¹ Information access

² Core Data Services

Standard development flow



SUMMARY



Key takeaways

ABAP Core Data Services (CDS)

Code pushdown is the ABAP programming paradigm on SAP HANA

CDS offers advanced capabilities to fully leverage SAP HANA's power in ABAP applications

CDS is the next generation data modelling infrastructure on the ABAP platform

CDS is the cornerstone of the ABAP RESTful Application Programming Model (RAP)

CDS plays a vital role in SAP BTP ABAP environment and SAP S/4HANA on-premise as well as in the cloud

Further enhancements and optimizations will be delivered with future ABAP platform releases

More information

SAP Community

- ❖ [ABAP Development Community](#)
- ❖ Landing pages: [ABAP Platform 2021](#) | [ABAP Platform 2020](#) | [ABAP Platform 1909](#) | [ABAP Platform 1809](#) | [AS ABAP 7.52](#) | [AS ABAP 7.51](#) | [AS ABAP 7.50](#)
- ❖ [Getting started with ABAP development for SAP HANA](#)
- ❖ [Getting started with ABAP Core Data Services \(CDS\)](#)
- ❖ [Getting started with the ABAP RESTful Application Programming Model](#)
- ❖ Developer center: <https://www.sap.com/developer/topics/abap-platform.html>

SAP Help Portal

- ❖ [ABAP Platform 2021](#) | [ABAP CDS Developer Guide](#) | [ABAP Keyword Documentation](#)

SAP training and certification opportunities

- ❖ www.sap.com/education

Thank you.